🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.7

Reflect Points Across Axes

Lesson 7-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.
Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Reflect Points Across Axes | **Reflection** | **x-axis** | **y-axis** | **Symmetry** | **Integer** | **Perpendicular**

👆 *Tap any word to see what it means and a picture.*

1. ____________ — Flip a point across an axis so it stays the same perpendicular distance from that axis.

2. A flip of a point or shape over an axis to make a mirror image is a ____________.

3. The horizontal number line on the coordinate plane is the ____________.

4. The vertical number line on the coordinate plane is the ____________.

5 When two halves are mirror images of each other, the figure has ________.

6 A whole number or its opposite, with no fraction part, is an ________.

7 Two lines that meet at a 90° square corner are ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How do reflections help the game designer create a balanced, symmetric level?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Reflect Points Across Axes** — Flip a point across an axis so it stays the same perpendicular distance from that axis.
  Reflejar puntos sobre los ejes — Voltear un punto sobre un eje para que quede a la misma distancia perpendicular de ese eje.
- [ ] **Reflection** — A flipped, mirror image across a line.
  Reflexión — Una imagen reflejada al otro lado de una línea.
- [ ] **x-axis** — The line that goes across the grid. Flipping over it changes the y sign.
  Eje x — La línea que va de lado en la cuadrícula. Voltear sobre ella cambia el signo de y.
- [ ] **y-axis** — The line that goes up the grid. Flipping over it changes the x sign.
  Eje y — La línea que va hacia arriba en la cuadrícula. Voltear sobre ella cambia el signo de x.
- [ ] **Symmetry** — When a shape looks the same on both sides of a line.
  Simetría — Cuando una figura se ve igual a ambos lados de una línea.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The enemy reflected over the y-axis is at ____ because ____. The enemy reflected over the x-axis is at ____ because ____. Reflections help game designers because ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Reflecting over the y-axis changes the ____ coordinate because ____.

Evidence: Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Reasoning: This evidence connects the definition of reflect points across axes to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The enemy reflected over the y-axis is at ____ because ____. The enemy reflected over the x-axis is at ____ because ____. Reflections help game designers because ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Reflect Points Across Axes
2. **Notes 2:** Reflection
3. **Notes 3:** x-axis
4. **Notes 4:** y-axis
5. **Notes 5:** Symmetry
6. **Notes 6:** Integer
7. **Notes 7:** Perpendicular
8. **Watch for this mistake:** *A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate — changing the x-coordinate when reflecting over the x-axis, instead of the y-coordinate (or vice versa for the y-axis). For example, to reflect (3, -5) over the x-axis, a student might flip the x-coordinate and write (-3, -5), but reflecting over the x-axis only changes the sign of the y-coordinate: (3, -5) becomes (3, 5). Before you submit, ask: 'Did I flip the sign of the coordinate that matches the axis I'm reflecting over — y for the x-axis, x for the y-axis?'*
9. **Try It 1:** A. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
10. **Try It 2:** A. The y-coordinate — *Across the x-axis the rule is (x, y) -> (x, -y), so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*